

ass5

March 2, 2025

```
[40]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, \
    confusion_matrix, roc_curve, auc
```

```
[42]: file_path = "C:/Users/cool/Desktop/ML ass/ass5/samples_cancer.csv"
cancer_data = pd.read_csv(file_path)
print("Dataset Preview:\n", cancer_data.head())
print("\nDataset Info:")
print(cancer_data.info())
print("\nMissing Values in Each Column:\n", cancer_data.isnull().sum())
print("\nChecking for '?' values:\n", cancer_data.isin(["?"]).sum())
```

Dataset Preview:

	ID	Clump	UnifSize	UnifShape	MargAdh	SingEpiSize	BareNuc	\
0	1000025	5	1	1	1	2	1	
1	1002945	5	4	4	5	7	10	
2	1015425	3	1	1	1	2	2	
3	1016277	6	8	8	1	3	4	
4	1017023	4	1	1	3	2	1	

	BlandChrom	NormNucl	Mit	Class
0	3	1	1	2
1	3	2	1	2
2	3	1	1	2
3	3	7	1	2
4	3	1	1	2

Dataset Info:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 699 entries, 0 to 698

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
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```

---      -----      -----      -----
0  ID          699 non-null    int64
1  Clump       699 non-null    int64
2  UnifSize    699 non-null    int64
3  UnifShape   699 non-null    int64
4  MargAdh     699 non-null    int64
5  SingEpiSize 699 non-null    int64
6  BareNuc     699 non-null    object
7  BlandChrom  699 non-null    int64
8  NormNucl    699 non-null    int64
9  Mit         699 non-null    int64
10 Class       699 non-null    int64
dtypes: int64(10), object(1)
memory usage: 60.2+ KB
None

```

Missing Values in Each Column:

```

ID          0
Clump       0
UnifSize    0
UnifShape   0
MargAdh     0
SingEpiSize 0
BareNuc     0
BlandChrom  0
NormNucl    0
Mit         0
Class       0
dtype: int64

```

Checking for '?' values:

```

ID          0
Clump       0
UnifSize    0
UnifShape   0
MargAdh     0
SingEpiSize 0
BareNuc     16
BlandChrom  0
NormNucl    0
Mit         0
Class       0
dtype: int64

```

```

[64]: cancer_data["Class"] = cancer_data["Class"].replace({2: 0, 4: 1})
      print("Unique values in Class column after conversion:", cancer_data["Class"].
      ↪unique())

```

```

cancer_data.replace("?", np.nan, inplace=True)
cancer_data = cancer_data.apply(pd.to_numeric, errors='coerce')
cancer_data.dropna(inplace=True)
cancer_data["Class"] = cancer_data["Class"].replace({2: 0, 4: 1})
print("Unique values in Class column:", cancer_data["Class"].unique())
print("\nUpdated Dataset Info After Cleaning:\n", cancer_data.info())

```

Unique values in Class column after conversion: [0 1]

Unique values in Class column: [0 1]

<class 'pandas.core.frame.DataFrame'>

Index: 683 entries, 0 to 698

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
0	ID	683 non-null	int64
1	Clump	683 non-null	int64
2	UnifSize	683 non-null	int64
3	UnifShape	683 non-null	int64
4	MargAdh	683 non-null	int64
5	SingEpiSize	683 non-null	int64
6	BareNuc	683 non-null	float64
7	BlandChrom	683 non-null	int64
8	NormNucl	683 non-null	int64
9	Mit	683 non-null	int64
10	Class	683 non-null	int64

dtypes: float64(1), int64(10)

memory usage: 64.0 KB

Updated Dataset Info After Cleaning:

None

```

[66]: X = cancer_data.drop(columns=["Class"]) # Features
      y = cancer_data["Class"] # Target variable (0 = Benign, 1 = Malignant)
      scaler = StandardScaler()
      X_scaled = scaler.fit_transform(X)
      X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2,
      random_state=42)

```

```

[48]: model = LogisticRegression()
      model.fit(X_train, y_train)
      y_pred = model.predict(X_test)
      y_pred_prob = model.predict_proba(X_test)[: , 1] # Get probability scores for
      ROC curve

```

```

[50]: accuracy = accuracy_score(y_test, y_pred)
      print(f"Model Accuracy: {accuracy:.4f}")
      print("\nClassification Report:\n", classification_report(y_test, y_pred))

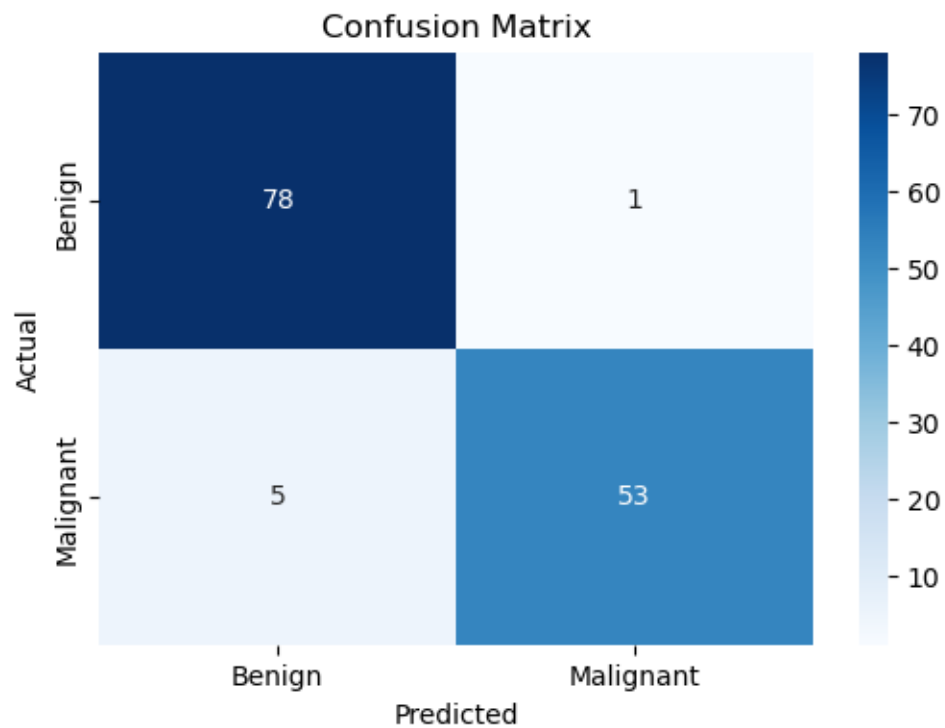
```

Model Accuracy: 0.9562

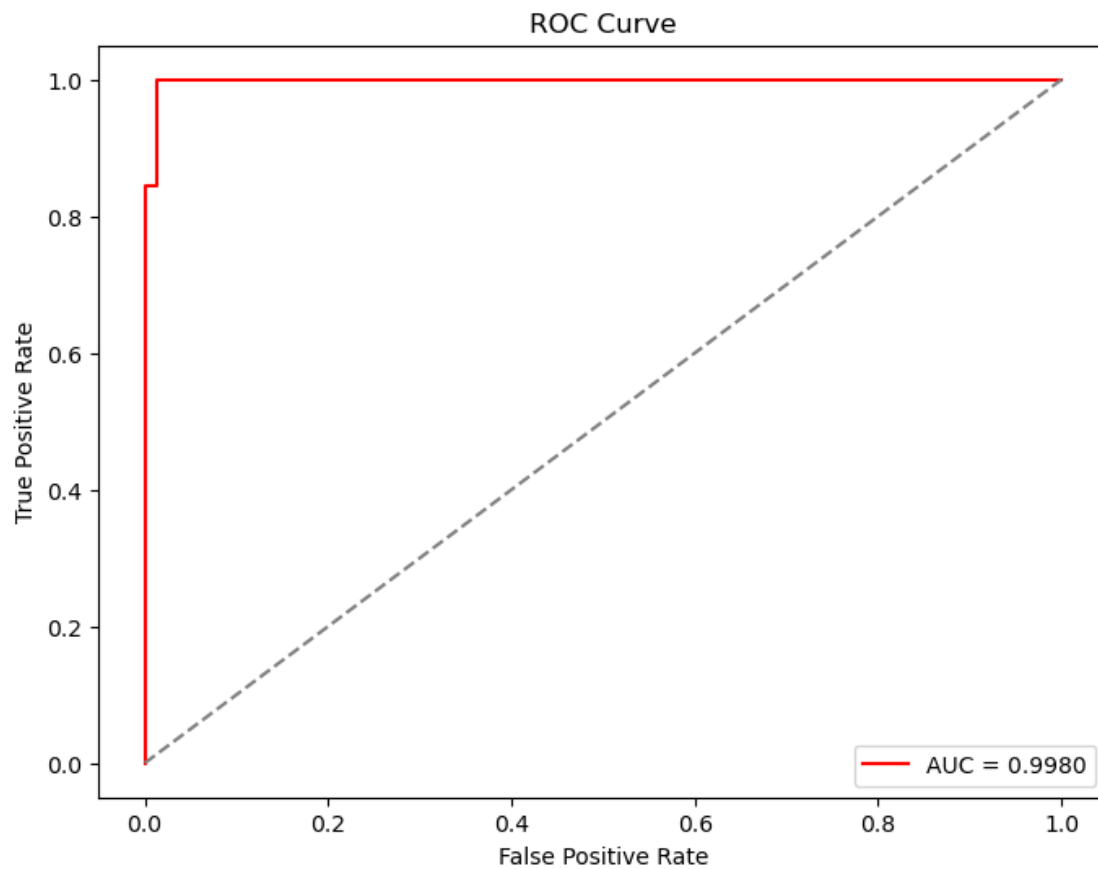
Classification Report:

	precision	recall	f1-score	support
2	0.94	0.99	0.96	79
4	0.98	0.91	0.95	58
accuracy			0.96	137
macro avg	0.96	0.95	0.95	137
weighted avg	0.96	0.96	0.96	137

```
[52]: cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=["Benign", "Malignant"],
            yticklabels=["Benign", "Malignant"])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
[68]: fpr, tpr, _ = roc_curve(y_test, y_pred_prob)
roc_auc = auc(fpr, tpr)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, color="red", label=f"AUC = {roc_auc:.4f}")
plt.plot([0,1], [0,1], linestyle="--", color="gray") # Baseline
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```



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[ ]:
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